

MATH 102 - Applied Linear Algebra

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Contents

1	Invertible Matrix Theorem	2
1.1	Definition	2
1.2	Thm 9.1.16	3
2	Annihilators	4
2.1	Theorem 9.3.3	4
2.1.1	Is The Converse True?	4
2.2	Each Matrix has an Eigenvector	5
2.2.1	Proof 1	5
2.2.2	Proof 2	5
3	Eigenvalues	5
3.1	Lemma 9.3.2	5
3.2	2-Commute	6
3.3	Gershgorin Disk Theorem	8
3.4	Corollary 9.4.15	8
4	Similar Matrices	9
4.1	Theorem 9.6.3	9

1 Invertible Matrix Theorem

1.1 Definition

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either all true or all false.

1. A is invertible.
2. A is row equivalent to I .
3. A has n pivot positions.
4. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
5. The columns of A are linearly independent.
6. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
7. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
8. The columns of A span $\mathbb{R}^n$.
9. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
10. There is an $n \times n$ matrix C such that $CA = I$.
11. There is an $n \times n$ matrix D such that $AD = I$.
12. A^T is invertible.
13. The columns of A form a basis of $\mathbb{R}^n$.
14. $\text{Col } A = \mathbb{R}^n$.
15. $\dim \text{Col } A = n$.
16. $\text{rank } A = n$.
17. $\text{Nul } A = \{0\}$.
18. $\dim \text{Nul } A = 0$.
19. The number 0 is *not* an eigenvalue of A .

- 20. $\det A \neq 0$.
- 21. $(\text{Col } A)^\perp = \mathbf{0}$
- 22. $(\text{Nul } A)^\perp = \mathbb{R}^n$
- 23. $\text{Row } A = \mathbb{R}^n$

1.2 Thm 9.1.16

Let $A \in M_n$ and let $\lambda \in \mathbb{C}$. Then the following statements are equivalent:

- (a) λ is an eigenvalue of A .
- (b) $A\mathbf{x} = \lambda\mathbf{x}$ for some nonzero $\mathbf{x} \in \mathbb{C}^n$.
- (c) $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, that is, $\text{nullity}(A - \lambda I) > 0$.
- (d) $\text{rank}(A - \lambda I) < n$.
- (e) $A - \lambda I$ is not invertible.
- (f) $A^\top - \lambda I$ is not invertible.
- (g) λ is an eigenvalue of $A^\top$.

Proof

- (a) $\Leftrightarrow$ (b)

By definition 9.1.1, if $A\mathbf{x} = \lambda\mathbf{x}$ and $\mathbf{x} \neq \mathbf{0}$ then $(\lambda, \mathbf{x})$ is an eigenpair of A , meaning λ is an eigenvalue of A .

- (b) $\Leftrightarrow$ (c)

These are restatements of one another.

- (c) $\Leftrightarrow$ (d)

By rank nullity theorem, if $\text{nullity}(A - \lambda I) > 0$, then $\text{rank}(A - \lambda I) < n$.

- (d) $\Leftrightarrow$ (e)

By property 1 and 16 of IMT, if $\text{rank}(A - \lambda I) < n$ then $(A - \lambda I)$ is not invertible.

(e) $\Leftrightarrow$ (f)

Suppose we have $A - \lambda I = \begin{bmatrix} a - \lambda I & b \\ c & d - \lambda I \end{bmatrix}$, and suppose it is not invertible, its determinant is $(a - \lambda I)(d - \lambda I) - bc = 0$. Consider the case of $A^\top - \lambda I$, we have $\begin{bmatrix} a - \lambda I & c \\ b & d - \lambda I \end{bmatrix}$, whose determinant is $(a - \lambda I)(d - \lambda I) - cb$. And by commutativity of multiplication, $bc = cb$, so the determinant is also 0, meaning $A^\top - \lambda I$ is also not invertible.

(f) $\Leftrightarrow$ (g)

Since (a) is equivalent to (e), the same is true for $A^\top$.

2 Annihilators

2.1 Theorem 9.3.3

Let $A \in \mathbb{M}_n$ and let p be a nonconstant polynomial that annihilates A . Every eigenvalue of A is a root of $p(z) = 0$.

Proof:

Let $(\lambda, \mathbf{x})$ be an eigenpair of A . Then $\mathbf{0} = 0_n \mathbf{x} = p(A) \mathbf{x} = p(\lambda) \mathbf{x}$, so $p(\lambda) \mathbf{x} = \mathbf{0}$

2.1.1 Is The Converse True?

While all the eigenvalues of a matrix are roots of each of its annihilators, the roots of an annihilator are not necessarily eigenvalues.

This raises a question:

Must some of the roots of an annihilator of a matrix be eigenvalues of the matrix? The answer is yes.

Let $p(z) = a_0 + a_1 z + a_2 z^2 + \dots + a_k z^k$ be an annihilator of a matrix $A \in \mathbb{M}_n$. Factor p as follows: $p(z) = a_k(z - \lambda_1)(z - \lambda_2) \dots (z - \lambda_k)$, $a_k \neq 0$. Since p annihilates A , we have

$$0 = p(A) = a_k(A - \lambda_1 I)(A - \lambda_2 I) \dots (A - \lambda_k I)$$

Since the zero matrix is not invertible, one of the factors $A - \lambda_i I$ is not invertible. This ensures that λ_i is an eigenvalue of A .

2.2 Each Matrix has an Eigenvector

2.2.1 Proof 1

λ is an eigenvalue of $A \in \mathbb{M}_n$ iff $\det(A - \lambda I) = 0$.

$p(\lambda) = \det(A - \lambda I)$ is a polynomial of degree n (the characteristic polynomial of A).

Therefore, λ is an eigenvalue of A iff λ is a root of p (i.e. p)

By the Fundamental Theorem of Algebra (FTA), each polynomial over the complex has a root.

Hence, each matrix $A \in \mathbb{M}_n$ has an eigenvalue.

2.2.2 Proof 2

We've shown that each matrix $A \in \mathbb{M}_n$ has an annihilator p .

By the FTA, p must have a root.

In 2.1, we showed that at least one root of p is an eigenvalue of A .

Hence, each matrix $A \in \mathbb{M}_n$ has an eigenvalue.

3 Eigenvalues

3.1 Lemma 9.3.2

Let (λ, x) be an eigenpair of $A \in \mathbb{M}_n$ and let p be a polynomial. Then $p(A)x = p(\lambda)x$, that is, $(p(\lambda), x)$ is an eigenpair of $p(A)$.

Proof:

Recall $A^k \mathbf{x} = \lambda^k \mathbf{x}$.

$$\begin{aligned} p(A)\mathbf{x} &= (c_k A^k + c_{k-1} A^{k-1} + \dots + c_1 A + c_0 I)\mathbf{x} \\ &= c_k A^k \mathbf{x} + c_{k-1} A^{k-1} \mathbf{x} + \dots + c_1 A \mathbf{x} + c_0 I \mathbf{x} \\ &= c_k \lambda^k \mathbf{x} + c_{k-1} \lambda^{k-1} \mathbf{x} + \dots + c_1 \lambda \mathbf{x} + c_0 \mathbf{x} \\ &= (c_k \lambda^k + c_{k-1} \lambda^{k-1} + \dots + c_1 \lambda + c_0) \mathbf{x} \\ &= p(\lambda) \mathbf{x} \end{aligned}$$

3.2 2-Commute

Let $A_1, A_2 \in \mathbb{M}_n$. Let λ be an eigenvalue of A_1 . Suppose $A_1 A_2 = A_2 A_1$, then A_1, A_2 share an eigenvector in $\mathcal{E}_\lambda(A_1)$

Computation:

0. Choose A_1 or A_2 . Say, A_1
1. Compute an eigenvalue of A_1 . Say, λ , and form its eigenspace $\mathcal{E}_\lambda(A_1)$
2. Select a basis $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_k$ for $\mathcal{E}_\lambda(A_1)$ and form a matrix $X = [\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_k]$
3. Solve $A_2 X = X C$ for C
4. Find an eigenpair of $C : (\mu, u)$
5. Compute $v = Xu$

Proof:

$$Ax_1 = \lambda x_1, \quad Ax_2 = \lambda x_2, \quad \dots, \quad Ax_k = \lambda x_k$$

$$A\vec{x}_i = \lambda \vec{x}_i \quad \text{for } i = 1, 2, \dots, k$$

$$A_1 x_i = \lambda x_i$$

$$A_2(A_1 x_i) = A_2(\lambda x_i)$$

$$A_1(A_2 x_i) = \lambda(A_2 x_i) \quad (\text{commutativity})$$

Case 1: $A_2 x_i = 0$
 $\Rightarrow A_2 x_i \in \mathcal{E}_\lambda(A_1)$

Case 2: $A_2 x_i \neq 0$
 $\Rightarrow A_2 x_i$ is an eigenvector of A_1 and $A_2 x_i \in \mathcal{E}_\lambda(A_1)$

$$A_2 x_1 = X c_1, \quad A_2 x_2 = X c_2, \quad \dots, \quad A_2 x_k = X c_k$$

$$A_2 [x_1 \ x_2 \ \dots \ x_k] = [A_2 x_1 \ A_2 x_2 \ \dots \ A_2 x_k] = [X c_1 \ X c_2 \ \dots \ X c_k] = X [c_1 \ c_2 \ \dots \ c_k]$$

$$A_2 X = X C$$

Let (μ, u) be an eigenpair of C $C\vec{u} = \mu\vec{u}$ $u \neq 0$

Say $v = Xu \neq 0$

$$Cu = \mu u$$

$$XCu = X\mu u$$

$$A_2 Xu = \mu Xu$$

$$A_2 v = \mu v$$

$\therefore v = Xu \in \mathcal{E}_\lambda(A_1)$ is an eigenvector of A_2

3.3 Gershgorin Disk Theorem

If $n \geq 2$ and $A = [a_{ij}] \in \mathbb{M}_n$ then

$$\text{spec } A \subseteq \mathcal{G}(A) = \bigcup_{k=1}^n \mathcal{G}_k(A).$$

Proof:

Let $(\lambda, \mathbf{x})$ be an eigenpair of $A = [a_{ij}] \in \mathbb{M}_n$ and let $\mathbf{x} = [x_i]$. Since $\lambda \mathbf{x} = A\mathbf{x}$, we have

$$\begin{aligned} \lambda x_i &= \sum_{j=1}^n a_{ij} x_j = a_{ii} x_i + \sum_{j \neq i} a_{ij} x_j \\ (\lambda - a_{ii}) x_i &= \sum_{j \neq i} a_{ij} x_j \end{aligned}$$

Let $k \in \{1, 2, \dots, n\}$ be any index such that $|x_k| = \max_{1 \leq i \leq n} |x_i|$. Since $\mathbf{x} \neq \mathbf{0}$, it follows that $|x_k| > 0$ and $\frac{|x_i|}{|x_k|} \leq 1$ for all $i = 1, 2, \dots, n$. Set $i = k$ and divide by x_k to obtain

$$\lambda - a_{kk} = \sum_{j \neq k} a_{kj} \frac{x_j}{x_k}$$

By the triangle inequality,

$$|\lambda - a_{kk}| = \left| \sum_{j \neq k} a_{kj} \frac{x_j}{x_k} \right| \leq \sum_{j \neq k} |a_{kj}| \left| \frac{x_j}{x_k} \right| \leq \sum_{j \neq k} |a_{kj}| = R'_k(A)$$

Therefore, $\lambda \in \mathcal{G}_k(A) \subseteq \mathcal{G}(A)$

3.4 Corollary 9.4.15

Let $A \in \mathbb{M}_n$. If A is strictly diagonally dominant, then it is invertible.

Proof:

The strict diagonal dominance of A ensures that $0 \notin \mathcal{G}_k(A)$ for each $k = 1, 2, \dots, n$. The Gershgorin Disk Theorem then tells us that 0 is not an eigenvalue of A and therefore A is invertible.

4 Similar Matrices

4.1 Theorem 9.6.3

If two real matrices are similar, then they are similar via a real matrix.

Proof:

Let $A, B \in \mathbb{M}_n(\mathbb{R})$. Let $S \in \mathbb{M}_n$ be invertible such that $A = SBS^{-1}$. For any $\theta \in (-\pi, \pi]$, let $S_\theta = e^{i\theta}S$ and compute $A = (e^{i\theta}S)B(e^{-i\theta}S^{-1}) = S_\theta BS_\theta^{-1}$. Then $AS_\theta = S_\theta B$. The complex conjugate of this identity is $A\bar{S}_\theta = \bar{S}_\theta B$ since A and B are real. Let $R_\theta = S_\theta + \bar{S}_\theta = 2 \operatorname{Re} S_\theta$ which is real for all $\theta \in (-\pi, \pi]$. Then $AR_\theta = A(S_\theta + \bar{S}_\theta) = (S_\theta + \bar{S}_\theta)B = R_\theta B$. The computation

$$R_\theta = S_\theta(I + S_\theta^{-1}\bar{S}_\theta) = S_\theta(I + e^{-2i\theta}S^{-1}\bar{S}) = e^{-2i\theta}S_\theta(e^{2i\theta}I + S^{-1}\bar{S})$$

shows that R_θ is invertible if $-e^{-2i\theta}$ is not an eigenvalue of $S^{-1}\bar{S}$. Since $S^{-1}\bar{S}$ has at most n distinct eigenvalues, there is a ϕ such that $-e^{-2i\phi}$ is not an eigenvalue of $S^{-1}\bar{S}$. Then $AR_\phi = R_\phi B$, where R_ϕ is real and invertible, and $A = R_\phi BR_\phi^{-1}$.